

weights across all tokens for input x_i . To determine which experts are included, we first rank experts based on their routing weight and then select top-ranked experts until their cumulative routing weight exceeds a predefined threshold. The weight threshold selects the experts that actually drive each input’s computation, rather than picking a fixed number of experts. In our main analysis, we use a cumulative routing-weight threshold of 0.7, and Appendix E.1 shows that the observed pathway trends remain consistent when varying this threshold. The selected experts are then concatenated as a comma-separated string (e.g., ‘3,1,5’) which are later joined across layers with hyphens (e.g., ‘3,1,5-...-9,1’).

We use the Levenshtein edit distance $D_{\text{path}}(s_i, s_j) = \text{EditDistance}(s_i, s_j)$, which counts the minimum insertions, deletions, or substitutions required to align two pathways, to measure pathway similarity. This metric captures both local deviations in expert choice and global shifts in pathway structure, reflecting differences in selection, length, and ordering. We monitor the average pairwise distance across samples throughout pretraining to track the emergence of shared routing.

Overall pattern. Our analysis, shown in Figure 4, reveals a clear trajectory in pathway dynamics. Early in pretraining, most inputs share nearly identical routes, producing low edit distance D_{path} . As the model memorizes individual inputs, pathways diverge and D_{path} rises. Crucially, D_{path} decreases after memorization, which indicates that semantically related inputs then converge into similar routing sequences, signaling the **emergence of shared pathways** that support generalization.

Layerwise pattern. While prior work [17] reports static depth-dependent specialization in trained MoEs, our results show that the memorization-to-generalization transition is also dynamically depth-dependent during pretraining (Figure 5). Shallow layers show the steepest decline in D_{path} after memorization, indicating rapid convergence to shared pathways. Deeper layers exhibit smaller reductions, and the final layer even shows an increase in D_{path} post-memorization. These patterns suggest a depth-specific reorganization: early layers consolidate universal representations, while later layers retain or enhance routing flexibility, balancing shared computation with task-specific specialization.

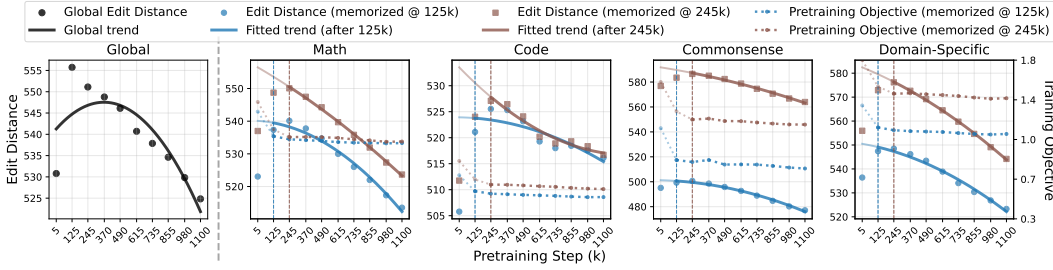


Figure 4: **Pathway edit distance and pretraining objective of two data groups** (memorized at 125k and 245k steps) during pretraining across four domains. The global panel (leftmost) fits the overall quadratic trend over all domains, while domain panels fit separate trends after the corresponding convergence step. Despite early training objective plateauing, pathway edit distance continues to decline, indicating a declining complexity of internal memorization.

4.2 Pathway Consistency and Complexity for Single Samples

We next analyze how the routing complexity of individual inputs evolves during pretraining. Our hypothesis is that increasingly streamlined and consistent expert selection across layers reflects the reuse of shared pathways, signaling the emergence of generalizable processing strategies. To capture this, we define **single-sample pathway consistency**, which measures the smoothness of expert transitions across consecutive layers.

In a typical MoE, the router at layer ℓ is a single-layer MLP—a linear transformation followed by a softmax—producing scores over K experts, with weight matrix $\mathbf{W}_\ell \in \mathbb{R}^{K \times d}$. We define the *expert embedding* for expert k as $\mathbf{v}_\ell^k := \mathbf{W}_\ell[k], k = 1, \dots, K$, which serves as an anchor vector to compute routing scores. For input x_i , its *routing embedding* is the weighted sum:

$$\mathbf{e}_\ell^{(i)} = \sum_{k=1}^K g_\ell^{(i,k)} \cdot \mathbf{v}_\ell^k,$$

where $g_\ell^{(i,k)}$ is the routing weight assigned to expert k . Although experts are independently parameterized, the strong correlations between adjacent layers in deep networks [8] and MoEs [14] suggest

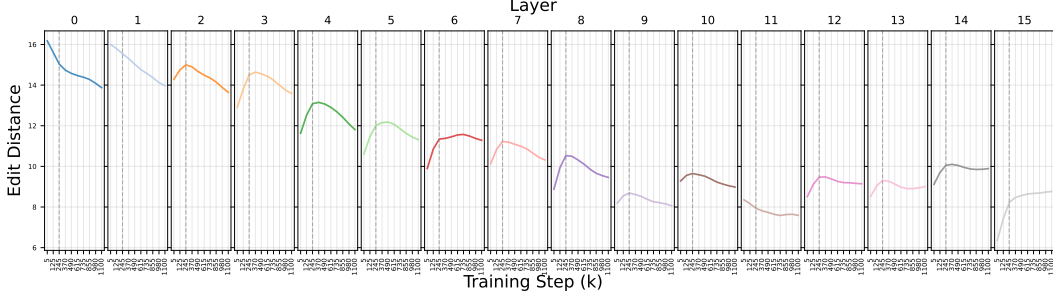


Figure 5: **Layer-wise evolution of pathway edit distance during pretraining** for the data group memorized at 245k step. Overall distance decreases from early to later layers, indicating higher complexity in early layers. After pretraining objective convergence, early-layer routing among similar samples simplifies quickly, later layers adapt more gradually, and the final layer diversifies after memorization.

that $\{\mathbf{v}_\ell^k\}$ can be comparable across neighboring layers, supporting structural analysis of routing behavior.

To quantify routing smoothness, we define **pathway consistency** C_i as the normalized average cosine similarity between consecutive layer embeddings for input x_i :

$$C_i = 1 - \frac{1}{L-1} \sum_{\ell=1}^{L-1} \frac{\cos(\mathbf{e}_{i,\ell}, \mathbf{e}_{i,\ell+1})}{\max_{\ell} \cos(\mathbf{e}_{i,\ell}, \mathbf{e}_{i,\ell+1}) + \epsilon},$$

where $\epsilon = 10^{-8}$ ensures numerical stability. Higher C_i indicates more erratic transitions, while lower values reflect smoother, more consistent routing across layers.

Figure 6 tracks the evolution of pathway consistency and pretraining objective across four domains. A key finding is that even after the objective stabilizes, pathway consistency continues to improve—revealing ongoing refinement in the internal routing of the model. This suggests that smoother, more coherent cross-layer transitions emerge well after memorization, positioning pathway consistency as a valuable lens for understanding generalization beyond what the training objective alone captures.

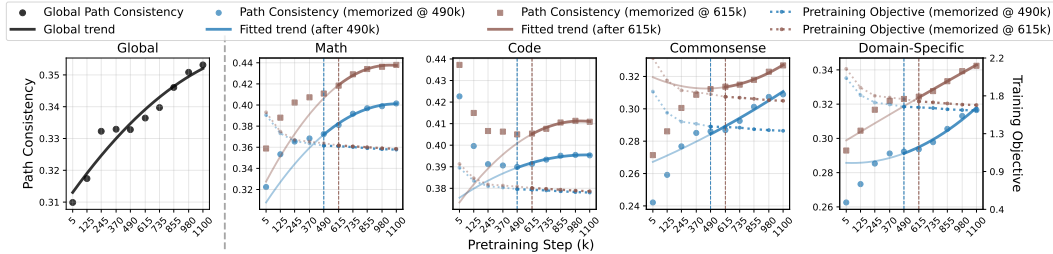


Figure 6: **Pathway consistency vs. training objective** on four domains during pretraining. The global panel (leftmost) shows that the path consistency consistently improves over all data, while each domain panel shows the same trend on two data groups (memorized at 490k and 615k) from each domain after the training objective converges. This reflects a progressively smoother transition of experts between consecutive layers after memorization.

4.3 Pathway Complexity Metrics as Monitors of Generalization

Given the strong link between routing dynamics and generalization, we ask: *To what extent do our pathway metrics predict test performance across domains?* To answer this, we compute Pearson and Spearman correlations between downstream performance and two metrics across pretraining checkpoints. As described in Section 3.1, we apply a lightweight LoRA-based instruction tuning (rank = 32; see Appendix B) solely to equip pretrained checkpoints with basic instruction-following

Table 1: **Correlation between pathway complexity metrics and generalization** (test accuracy) across four domains, with the comparison to training/validation loss. p -value: $p < 0.1$, $p < 0.05$,

$p < 0.01$, $p < 0.001$, $p > 0.1$, $p > 0.5$, $p > 0.8$. Preferred correlation: positive , negative .

Metric	Math		Code		Commonsense		Domain-Specific	
	Pearson	Spearman	Pearson	Spearman	Pearson	Spearman	Pearson	Spearman
Pathway Distance (pairwise)	-0.9471	-0.9487	-0.9290	-0.9276	-0.9821	-0.9747	-0.9254	-0.9487
Pathway Consistency (sample-wise)	0.9246	0.9487	0.9786	0.9856	0.9804	0.9747	0.9917	0.9487
Training Loss	-0.0435	-0.2108	-0.0680	-0.2052	0.6427	0.4104	0.7944	0.9487
Training Loss (moving average)	0.4714	0.3162	0.2481	-0.2052	0.7931	0.6669	0.8705	0.9487
Validation Loss	0.4801	-0.1054	0.5466	0.8721	-0.5752	-0.4617	-0.3339	-0.3162

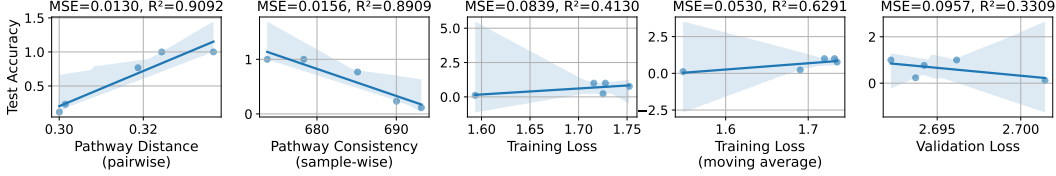


Figure 7: Correlation of the two pathway complexity metrics with benchmark performance. Compared to the training objective and its moving average, both the pathway edit distance and consistency are highly correlated with the test accuracy, though they are also computed on training data.

ability for benchmark evaluation. The robustness of this setup is further supported by the LoRA rank ablation in Appendix E.2. Our correlation analysis focuses on checkpoints after domain-level accuracy begins its sustained rise (shown in Figure 1), thereby isolating the mechanisms that drive generalization beyond the random-accuracy phase.

Table 1 summarizes the correlation results across four tasks. Pathway consistency (sample-wise) shows exceptionally strong positive correlations with test accuracy—often exceeding 0.97 and highly significant—indicating that coherent, stable routing across layers is a key marker of generalization. In contrast, pairwise pathway similarity consistently yields strong negative correlations (around -0.93), suggesting that as generalization improves, inputs tend to follow increasingly similar expert routes. By comparison, as shown in Figure 7, training and validation loss metrics exhibit weaker and less consistent correlations. Notably, while moving-average training loss shows relatively high correlation in certain domains (e.g., Code, Domain-Specific), its direction is inconsistent with expectation: since loss is minimized during training, we would expect a *negative* correlation with test accuracy—not the *positive* one observed here. This mismatch further underscores that conventional pretraining metrics fail to reflect the internal transition from memorization to generalization.

These findings position pathway metrics as robust, domain-general indicators of generalization. Importantly, they rely only on training dynamics—without requiring held-out validation data—making them especially valuable for real-time monitoring in large-scale or resource-constrained settings.

4.4 Theoretical Connections & Explanations

We connect expert routing patterns with model generalization via the *complexity* of routing, formalized through the *effective dimension* of the routing kernel. This effective dimension will serve as the key determinant of MoE generalization.

Concretely, $\mathbf{H}^*$ is the Gram matrix of the routing kernel with entries $\mathbf{H}_{ij}^* = \Theta_{\text{route}}(x_i, x_j)$, capturing similarity between inputs induced by both routing overlap and expert functions. The effective dimension is defined as $\text{Tr}(\mathbf{H}^*(\mathbf{H}^* + \lambda\mathbf{I})^{-1})$, measuring the complexity of the function space induced by routing.

Theorem 4.1 (Generalization Bound). *Under NTK regime with K experts, with probability $1 - \delta$ over n samples, where C_1, C_2 are NTK-dependent constants and λ is the regularization parameter:*

$$\mathcal{E} \leq \underbrace{\frac{C_1 \lambda^2}{n} \mathbf{y}^\top (\mathbf{H}^* + \lambda \mathbf{I})^{-2} \mathbf{y}}_{\text{Bias}} + \underbrace{C_2 \left(\frac{\sigma^2 \text{Tr}(\mathbf{H}^*(\mathbf{H}^* + \lambda \mathbf{I})^{-1})}{n} + \frac{\sigma^2 \log(1/\delta)}{n} \right)}_{\text{Variance + Noise}}$$